

OMKAR SINGH

Software Engineering Intern · Full-Stack Developer · IIT Kharagpur
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EDUCATION

Indian Institute of Technology (IIT) Kharagpur

B.Tech – Industrial & Systems Engineering

Coursework: Data Structures & Algorithms, Process Design, Project Management, Optimization

2023 – Present

GPA: 7.56 / 10.0

Douglas Memorial Higher Secondary School

Class 12: 80.6% | Class 10: 85%

EXPERIENCE

Software Engineering Virtual Experience

Aug – Sep 2025

J.P. Morgan Chase & Co. · Financial Technology · Remote

- Built an interactive financial data visualisation dashboard using **React.js** and **Python**-based APIs, improving data readability and reducing manual reporting effort.
- Refactored legacy data-processing logic and participated in **Git**-based code review workflows, simulating real-world **Agile** SDE team collaboration.

PROJECTS

Synthetic-Bull | Distributed Stock Trading Simulator

2025

Go (Gin/GORM) · MySQL · WebSockets · REST APIs · Microservices · Distributed Systems

- Architected 11-node distributed backend in **Go (Gin/GORM)** handling real-time order matching, portfolio management, and market data aggregation across decoupled microservices.
- Implemented custom Limit Order Book with Price-Time Priority — **O(log N)** insertion via Red-Black Trees, **O(1)** best-price retrieval, supporting partial fills.
- Engineered lock-free fan-out via **goroutines/sync.WaitGroup** broadcasting to 5 services at sub-ms latency; **REST** API Gateway + **WebSockets** for real-time tick-by-tick candlestick chart rendering.
- Built GBM market simulator generating 50–100 statistically realistic orders/sec, driving live OHLCV candlestick aggregation.

Collaborative Whiteboard | Real-Time Web App

Jul – Aug 2025

React.js · Firebase Firestore · HTML5 Canvas API · Tailwind CSS · Rough.js

- Architected multi-user collaborative drawing app with sub-100ms sync latency using **Firebase Firestore** listeners to propagate canvas state across clients.
- Built stateful drawing engine supporting 6+ tools with undo/redo history stack via **Canvas API** and **Rough.js**; designed mobile-first UI with **React Hooks**.

MERN E-Commerce Platform | Full-Stack Web Application

2025

MongoDB · Express.js · React.js · Node.js · JWT · Multer · Netlify

- Built a full-stack MERN e-commerce platform featuring **JWT** authentication with **HTTP-only cookies** (XSS-safe), role-based admin controls, and product image upload via **Multer**.
- Designed a **RESTful API** using **Express.js** and **Mongoose** with async error-handling middleware, bcrypt password hashing, and protected route guards.
- Implemented shopping cart, order management, product **CRUD**, and user reviews; configured concurrent dev workflow with **concurrently** and deployed React SPA to **Netlify**.

TECHNICAL SKILLS

Node.js · Express.js · MySQL · MongoDB · React.js · Python · Docker · Java

CERTIFICATIONS

J.P. Morgan SWE Virtual Externship

JPMorganChase

Apr – May 2026

The Web Developer Bootcamp 2025

Udemy

Jun 2025

Advanced Software Engineering Simulation

Walmart Global Tech (Forage)

Sep 2025

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

- LeetCode**: Solved 250+ problems — solved various problems on DSA
- Codeforces Rating**: 1286 (Specialist) — consistent performance in algorithmic contests.
- HackerRank**: 5-Star Gold Badge in Problem Solving.